Wi-Fi Signals, Side Lobes, and Spy Antennas:
The Story of Energy Leakage

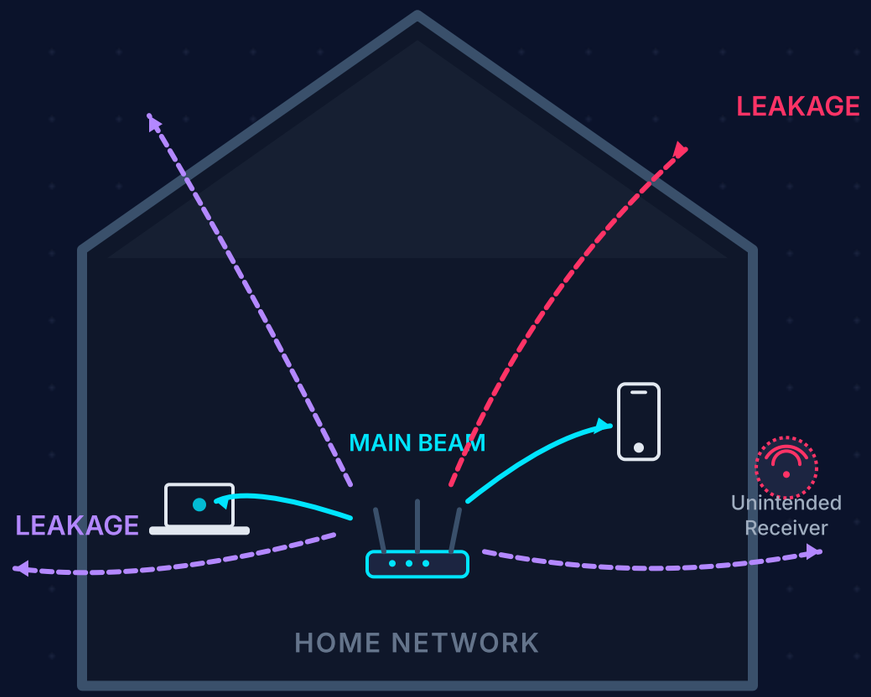
From Maxwell's equations to real-world signal leakage

Wi-Fi in the Real World

Wi-Fi is designed to provide seamless coverage for all our personal devices.

However, **some signal inevitably escapes** outside the boundaries of your home.

> This happens because standard antennas radiate energy in multiple directions, creating unavoidable "leakage."



Maxwell's Equations and Energy Flow

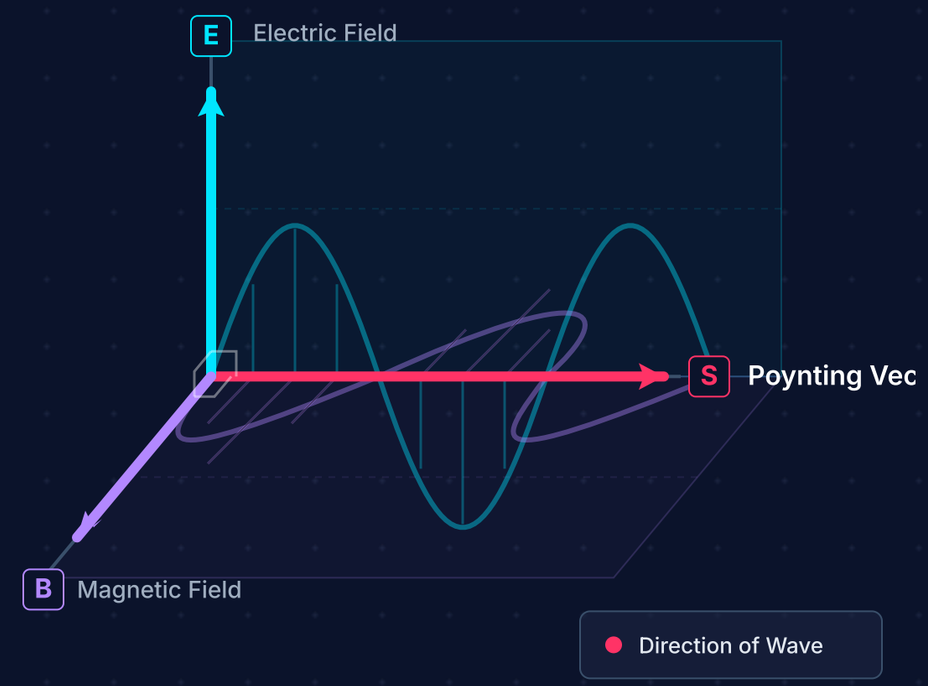
Electromagnetic waves (like Wi-Fi) are governed by **Maxwell's equations**, describing how electric and magnetic fields generate each other:

$$\nabla \cdot \mathbf{E} = \rho / \varepsilon_0$$

$$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 (\partial \mathbf{E} / \partial t)$$



The **Poynting vector** shows the direction and magnitude of energy flow:

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Antenna Arrays: Diffraction and Interference

A single antenna spreads energy broadly like a light bulb. Multiple antennas (an array) interfere with each other, creating peaks and valleys known as **main lobes** and **side lobes**.

$$\psi(\theta) = \psi_0 \times [\text{Diffraction}] \times [\text{Interference}]$$

$$[\ \sin(\pi a/\lambda \cdot \sin\theta) \ / \ (\pi a/\lambda \cdot \sin\theta)\ ]$$

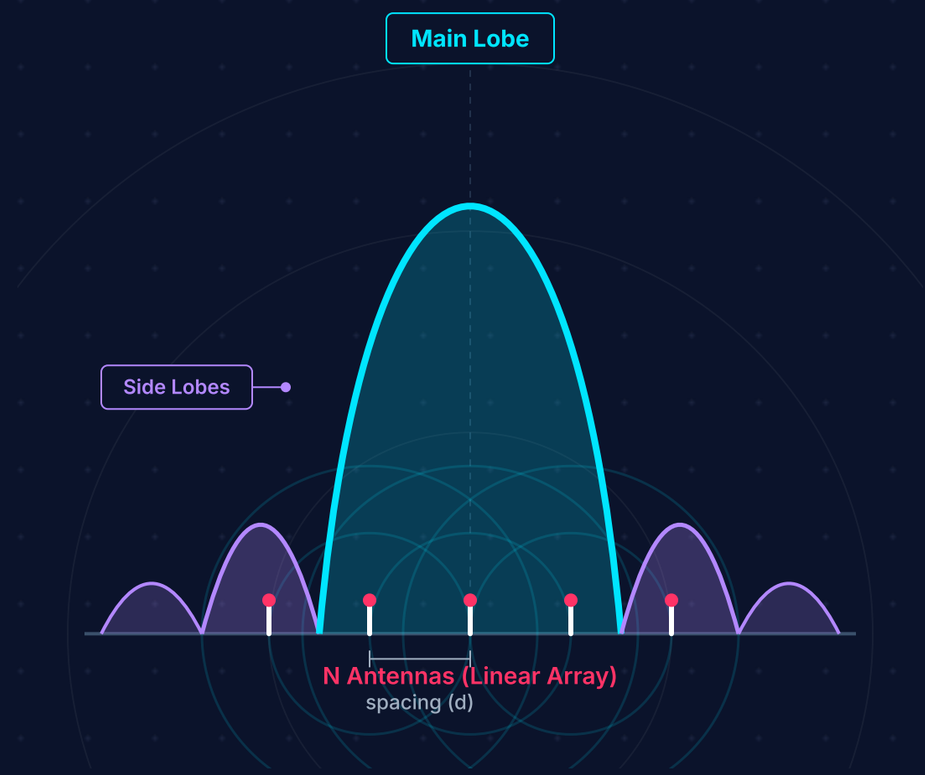
Term 1: Diffraction of a single antenna element

$$\times$$

$$[\ \sin((N/2)(kd \cdot \sin\theta + \varphi)) \ / \ \sin((kd/2)\sin\theta + \varphi)\ ]$$

Term 2: Array factor (Interference from N antennas)

The **first term** represents the natural spread of one antenna. The **second term** shapes the overall beam, where constructive interference forms the main peak and unavoidable side lobes.



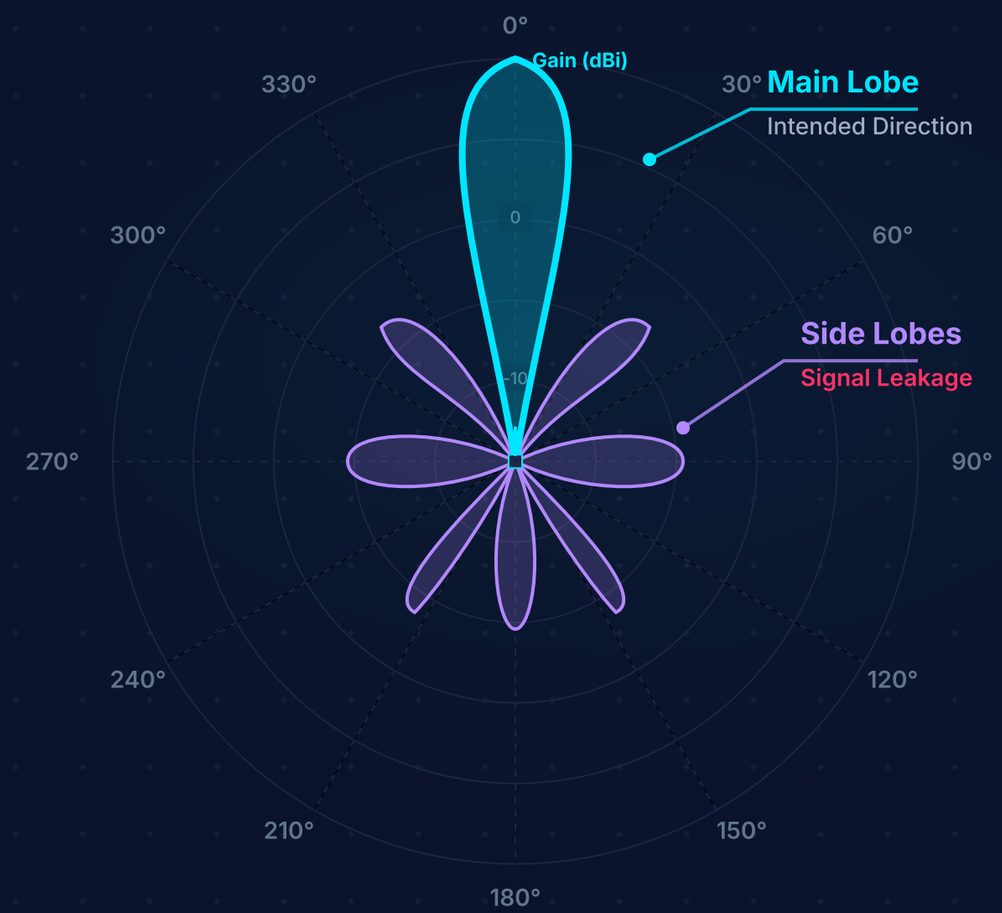
Intensity and Gain

The physical energy spreading from an antenna isn't uniform. The actual **Intensity** is the square of the electromagnetic field:

$$I(\theta) = |\psi(\theta)|^2$$

When interference shapes the wave, the intensity plot reveals distinct lobes:

- Main Lobe: The intended, strongest beam pointing toward your devices.
- Side Lobes: Smaller, unavoidable peaks that carry energy outside the main beam—this is how your signal leaks.



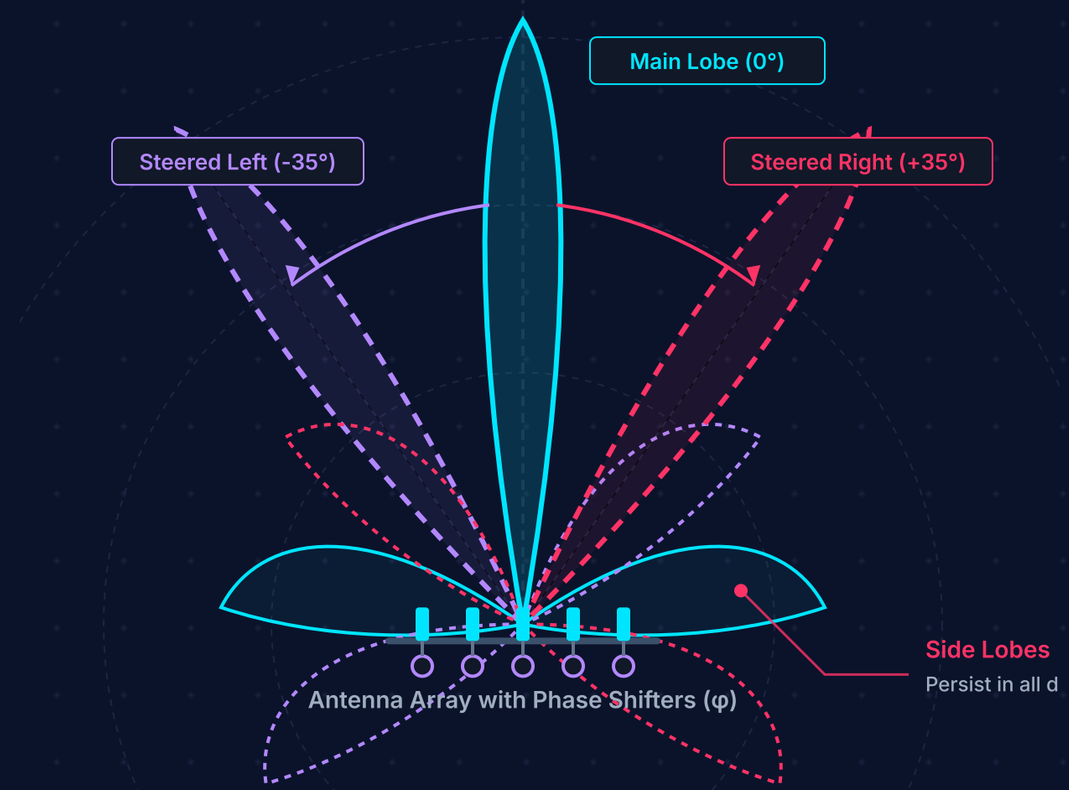
Steering the Beam: Phase Shifts (Beamforming)

By adjusting the phase φ of each individual antenna in an array, we manipulate the interference pattern to electronically "steer" the main beam.

STEERING ANGLE EQUATION

$$\theta_{max} \approx \sin^{-1}(-2\varphi / \pi)$$

While beamforming successfully directs the **Main Lobe** toward a target device, the **Side Lobes** simply rotate along with it. You can minimize their intensity, but they cannot be completely eliminated.



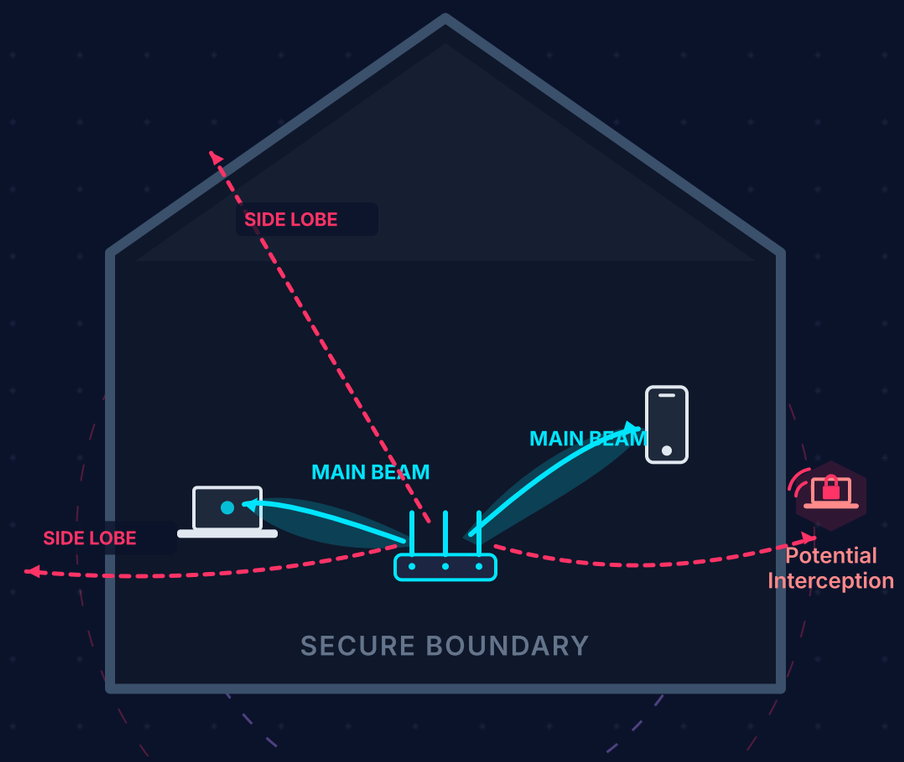
Real-World Example: Wi-Fi Router

Inside the home, modern routers use beamforming to direct a strong **main beam** toward your active devices (like laptops and phones).

However, outside the main beam, secondary energy peaks—known as **side lobes**—inevitably radiate through walls and windows.

SECURITY IMPLICATION

While beamforming improves targeted signal strength, it **cannot eliminate side lobes**. This leaked energy remains strong enough for potential signal interception outside the physical bounds of your home.



Summary: From Physics to Wi-Fi Security

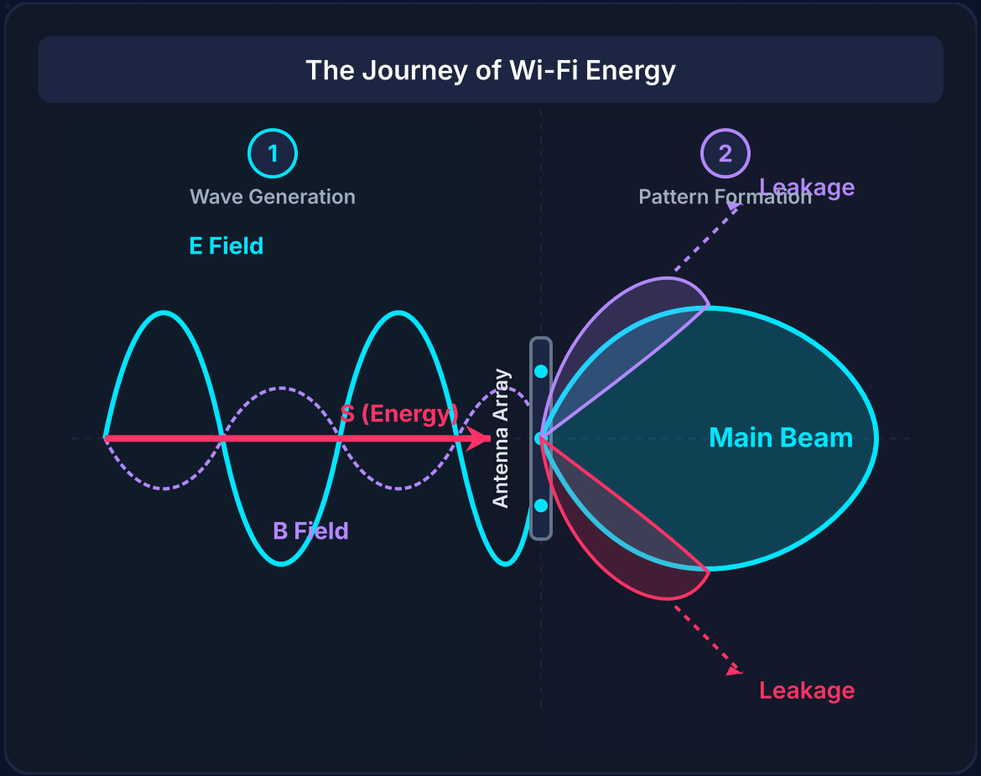
From fundamental physics to everyday devices, energy spreads in complex, predictable ways:

1. **Maxwell's Equations**
   Describe how E and B fields generate each other to create electromagnetic waves.

2. **Poynting Vector**
   Predicts the exact direction and magnitude of physical energy flow through space.

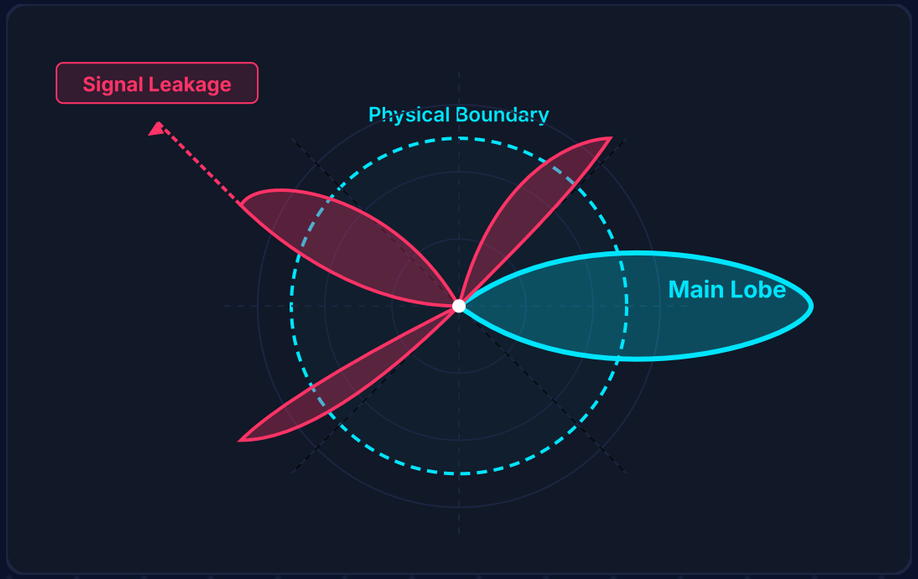
3. **Arrays & Interference**
   Multiple antennas shape the energy via interference, forming main and side lobes.

4. **Phase Shifts (Beamforming)**
   Allow us to steer the main beam, but cannot eliminate the surrounding leakage.

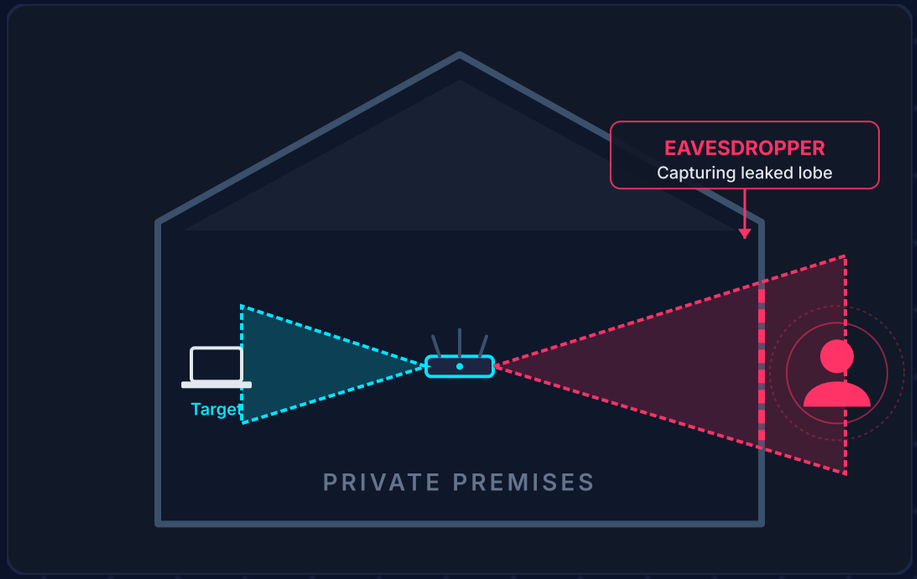


Bonus: Side Lobes as a Security Risk

Even in everyday Wi-Fi, **side lobes can leak sensitive information** beyond intended boundaries. Understanding these physical realities helps us design safer, more efficient networks.



THEORETICAL RADIATION PATTERN



REAL-WORLD VULNERABILITY